Supply Chain Management and Influencing Factors Based on E-commerce Delivery Data

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Abstract

This paper examines the impact of e-commerce on supply chain management by analyzing delivery times and generating insights for consumer preferences and product prioritization for suppliers. Utilizing datasets from a ten-day period in 2021, the research employs regression modeling and machine learning to evaluate and predict delivery performance across various factors. Insights derived from this analysis aim to optimize e-commerce operations and improve consumer experiences by identifying strategic improvements in delivery processes.

CCS Concepts

• **Applied computing** → Electronic commerce; E-commerce infrastructure.

Keywords

Delivery times, Consumer preferences, E-commerce operations

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1 Introduction

The advent of e-commerce has transformed the shopping habits of millions globally, making the monitoring of delivery times a critical aspect of the online shopping experience. Particularly in the aftermath of the COVID-19 pandemic, the e-commerce sector has witnessed unprecedented growth. Between 2019 and 2021, a period marked by worldwide disruptions due to the pandemic, e-commerce not only became a necessity for many but also underwent significant evolutionary changes in its operational dynamics. A noticeable surge in total global retail sales from 2019 to 2020 underscores this trend, reflecting a substantial increase in online transactions. This shift, catalyzed by the pandemic, has redefined consumer behavior and reshaped the business landscape, leading to a new paradigm in retail and supply chain management [1].

From the perspective of the e-commerce platforms, the "struggle of competing on price and shipping" is always one of the most common challenges. E-commerce businesses with different sizes can compete on the same kind of product, with larger and more prominent platforms can provide a cheaper price for the same good, with a more efficient delivery method. [2] Therefore, for some small and medium-sized platforms, how to formulate appropriate commodity prices and how to reasonably plan delivery methods and routes are the issues they need to consider. There is a critical need to scrutinize delivery timelines for various products based on their prioritization and utilize some models to provide consumers with appropriate estimated delivery time to bring a better experience.

For consumers, the focus shifts to identifying products that are consistently delivered on time or earlier, enabling them to make informed purchasing decisions. With more advanced e-commerce platforms and technology, the demand of consumers increases as well. They want a simple and more convenient shopping experience which is "personalized and engaging", being able to track the updated information of their orders.[3] Meanwhile, suppliers are tasked with prioritizing their products based on delivery performance. The reason is that since Covid-19, the inputs and availability of many materials became insufficient due to much higher consumer demands. To overcome this, suppliers need to optimize their manufacturing plan based on market research and construct strategic negotiations with e-commerce platforms to possibly evaluate and maybe elevate the priority of their products, thereby ensuring which products should be supplied more and ensuring better visibility and faster delivery of those products. [4]

This paper explores these dynamics within the context of supply chain management, investigating how the interplay between factors involving delivery times, product prioritization, and consumer preferences influences the efficacy of e-commerce operations. By analyzing delivery data from e-commerce transactions using RStudio and Python, this study aims to construct regression models to analyze which factors are important to predict and impact the delivery times of e-commerce products, and then uncover patterns and propose strategies that could enhance the efficiency and responsiveness of online shopping platforms, providing insights for platforms, customers, and also suppliers.

2 Data

2.1 Datasets

This study utilizes a collection of ten datasets corresponding to consecutive days from June 1, 2021, to June 10, 2021. Each dataset represents a full day's worth of e-commerce delivery data from various regions within Hong Kong. The datasets are comprehensive and standardized across all days, containing several consistent variables and categories that are crucial for the analysis.

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Table 1: Variables used in the model.

Style Tag	Definition
delivery_time	The time required from order date to delivery date
sub_cat_1_name_en	The product first-level subcategory
device_type	Indicates the type of device the order was made on
area	The area of delivery address
total_price_per_unit	The total price of each order
total_discounts	The range of total discounts
packing_box_type	The packing box type defined by internal
country	The country of the manufacturer
delivery_timeslot	Three time slots consumers can choose as a suitable delivery time window
membership_level	The membership level of the consumer
weekend	A binary variable depending on whether the order date is weekday or weekend

The variables included in these datasets encompass membership level, order date and time, device type, and geographical area. Additionally, each dataset details the subcategory of products ordered and the country of origin of these products. These variables are instrumental in examining the dynamics of e-commerce delivery within the stipulated period.

In the following sections, I will elaborate on the specific variables used in the modeling of this research, as well as the relevant data considerations that have informed the analytical framework of this study. This detailed examination aims to shed light on the factors influencing delivery times and customer satisfaction in the context of online shopping.

2.2 List of Variables

Table 1 is a list of the variables used in the regression model. The response variable is "delivery_time", and there are 10 explanatory variables with corresponding explanation. Regulatory checks, special packaging, handling requirements, and demand fluctuations of different categories of products all affect delivery times, often delaying products like alcoholic beverages and perishables while speeding up essentials. [5]

"country" can also be an important factor influencing the delivery time due to different locations and delivery routes, especially for e-commerce. Delivery times also vary with pricing strategies and consumer behavior obviously; higher-priced items often ship faster due to their value, while heavily discounted products may face delays due to higher demand and logistics prioritization during sales. [6] Additionally, due to different local postal services operating on weekends, weekdays, or public holidays, delivery time for different orders can also vary by the variable "weekend" and "delivery_timeslot". [7]

2.3 Overview of relationship between variables and delivery time

Figure 1 illustrates the relationship between categories of products and the delivery time, which is called "sub_cat_1_name_en". From this chart, we can see the average delivery time for SD (Snacks & Desserts) is the highest about 3 days, while the average delivery time for F_V_F (Fruits, Vegetables & Flowers) is the lowest about 1 day. The main reason could be the fruits, vegetables and flowers need to be preserved as quickly as possible. They need better preservation room than others. And products such as Seas (Seasonings), P_C (Personal Care), and R_CO (Rice & Cooking Oil) also take longer time to be delivered, which is also because of the property of these products.

Figure 2 is the chart about device type and corresponding delivery time. IOS appears to have a larger transaction count compared to Android and Web, indicating higher usage or sales through iOS devices. For Android and Web, these points are smaller than iOS, suggesting fewer transactions. IOS devices might be the most popular platform among users for making transactions, followed by Android and Web. IOS also seems to have the quickest average delivery time. This might indicate either a more streamlined processing for IOS transactions or possibly different types of products being ordered more frequently on IOS platforms that allow for quicker delivery. Known for its sleek and consistent interface, iOS often facilitates quicker and smoother e-commerce transactions, with users typically opting for faster, higher-tier delivery options, possibly due to factors like younger demographics, a preference for online shopping, higher average incomes, and job types compared to Android users.[8] Also, Web-based e-commerce is accessible on any device but depends heavily on internet speed and browser compatibility, which might slow down transaction processing.[9]

Also, as we can see from Figure 3, from the packing box type side, different packing box types correspond to totally different average delivery time. The materials and durability of packing boxes may impact their handling requirements, with more durable materials often processed faster due to reduced damage risk, while fragile boxes require more careful handling, potentially slowing delivery. Additionally, the size and weight of these boxes directly influence delivery times; larger or heavier boxes may need special handling, slowing down their transit, whereas smaller, lighter packages typically move through standard shipping processes more quickly, speeding up delivery (The classification of packing box types varies by region and platform. While the specific characteristics of each type are unclear, they play a crucial role in predicting delivery times).

In the following section I will dive into the model construction part and try to find more precise relationships between these variables and delivery time.

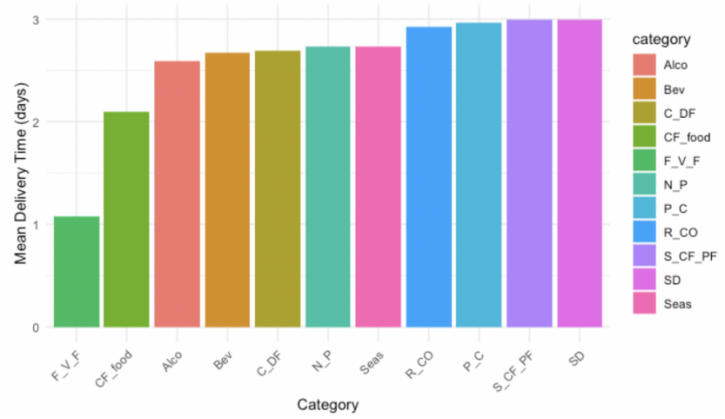

Figure 1: Average Delivery time by category.

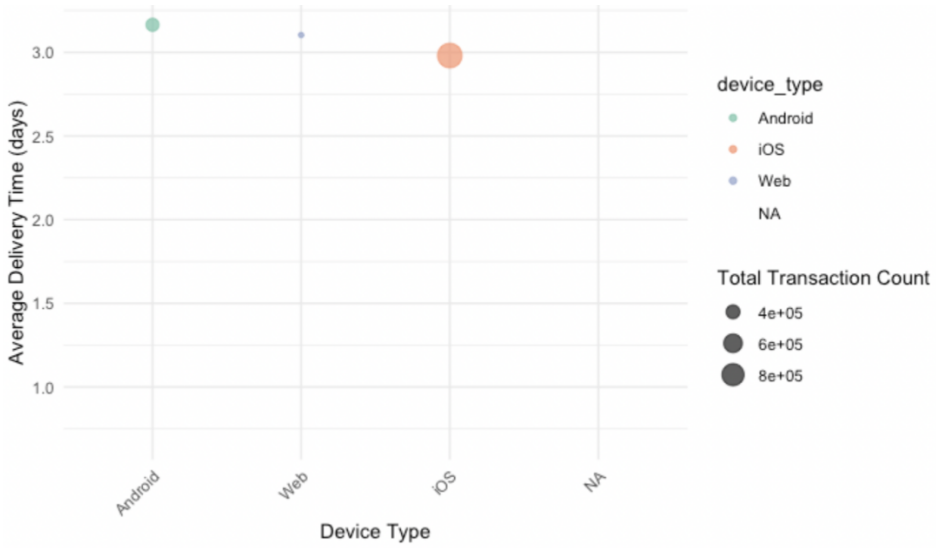

Figure 2: Average Delivery time by device type.

3 Regression Model

3.1 Model Settings

This study employs a multiple linear regression model in R to examine the impact of variables such as membership level, area, country, and packing box types on delivery times. I use the lm() function in R to model delivery time as a function of these variables, treating categorical variables like packaging types with the factor() function. The model will help determine how each factor affects delivery time (The response variable) and identify which are statistically significant, providing insights for optimizing e-commerce logistics to enhance customer satisfaction.

3.2 Modeling summary

For the reason of statistical significance, I selected all of the levels of variables which has a p-value smaller than 0.05, and the results of the modeling summary are shown in Table 2.

In a comprehensive analysis of e-commerce delivery times, various factors significantly impact how quickly products are delivered to consumers. Among these, product categories play a crucial role. Perishables and high-maintenance items such as Alcoholic Beverages, Chilled/Frozen Food, and Fruits, Vegetables & Flowers have longer delivery times compared to more straightforward items like accessories. On the other hand, categories such as Beach Gears & Water Activities and Balloons & Mahjong & Party Essentials see quicker deliveries, showcasing how product sensitivity and handling requirements influence logistical efficiency.

Device type and geographical location also markedly affect delivery speeds. Orders made from iOS devices tend to be processed faster than those from Android, possibly due to the streamlined nature of iOS's e-commerce systems and the spending habits of its user base. Within Hong Kong, variations in delivery times are evident across different districts; Hong Kong Island typically experiences faster deliveries compared to Kowloon and New Territories,

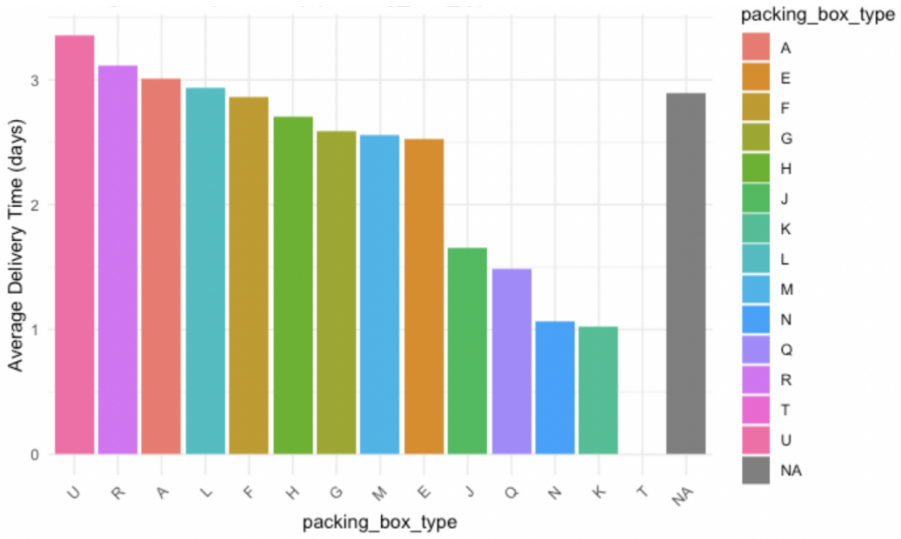

Figure 3: Average delivery time by packing box type.

Table 2: Results of the modeling summary.

Variables	Parameters	P-values
sub_cat_1_name_en	controlled	controlled
device_typeiOS	-0.2079	0.000000e+00
device_typeWeb	-0.0539	9.815680e-20
areaKLN	0.0592	7.093424e-30
areaNT	0.0673	5.122994e-48
total_price_per_unit	-0.0002	1.962369e-46
total_discounts	0.0012	0.000000e+00
delivery_timeslotevening	0.2031	0.000000e+00
delivery_timeslotmorning	0.4324	0.000000e+00
membership_levelNORMAL	-0.4119	0.000000e+00
membership_levelSENIOR	-0.3856	4.347814e-56
membership_levelVIP	-0.1597	0.000000e+00
packing_box_type	controlled	controlled
country	controlled	controlled

which can be attributed to differences in logistics infrastructure and population density.

Moreover, pricing strategies and membership levels significantly dictate delivery prioritization. High-priced items are often delivered faster, benefiting from priority shipping, while products with higher discounts may face delays, especially during peak sale periods due to increased demand. Membership tiers further influence delivery times, with gold VIP members often experiencing slower but higher-quality service, in contrast to Normal, Senior, and VIP levels, which enjoy faster delivery due to less complex service requirements. This layered analysis highlights the diverse factors that e-commerce companies must navigate to optimize delivery times effectively.

4 Machine learning model prediction and importance rank

After the analysis of regression model, the data is scaled using "StandardScaler" and been split into train sets and test sets to complete the machine learning task.

4.1 Random Forest

The Random Forest Regressor is trained on the already scaled data. There are 200 "trees" (estimators), and the maximum depth is set by 10 to control the selecting steps. Also, the threshold is set by 0.85 to select appropriate features. After selection procedure, there are ten features selected and the relative importance of each one is listed. "Area" has the highest importance score (0.607014), indicating it is the most significant predictor of delivery time of the model. The other important feature is "device type" (0.196800), also impacting

Table 3: Performance evaluation for Random Forest

	RMSE	R2	MAE
Training set	1.832587	0.229061	1.175508
Testing set	1.834341	0.222312	1.177021

Table 4: Performance evaluation for XGBoost Regressor.

	RMSE	R2	MAE
Training set	1.486685	0.492625	0.936022
Testing set	1.831462	0.224752	1.178544

Table 5: Importance Rank.

No.	Variables	Importance ratios
1)	area	0.3705
2)	device_type	0.1584
3)	packing_box_type	0.0709
4)	delivery_timeslot	0.0671
5)	total_discounts	0.0644
6)	weekend	0.0625
7)	membership_level	0.0549
8)	total_price_per_unit	0.0511
9)	sub_cat_1_name_en	0.0510
10)	country	0.0494

heavily on the prediction of the model. The least important feature selected is "country", which has the importance score just 0.004563, meaning it may have the minimum impact on model prediction.

The performance function is defined to evaluate the model by using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R2) through training set and testing set.

According to the result from Table 3, the RMSE and MAE are very similar for both the training set and testing set, indicating the model is not overfitting because the performance of two sets are not very different. The R2 values are quite low (just 0.229 for training set and 0.222 for testing set), meaning that the model only explains 22%-23% of the variance of delivery time, the response variable. This suggests the model is not very efficient at predicting the variable, and it needs to be improved by using more advanced and complex models to predict the variables, modifying threshold or adjusting hyperparameters when constructing model, etc.

4.2 XGBoost Regressor

Another training model is XGBoost Regressor, with max_depth equals 13, min_child_weight equals 5, and 200 estimators. The model is evaluated by the same performance metrics as above.

As shown in Table 4, the R2 value is higher on training set compared to testing set, meaning that the model explains nearly half of the variance in the training data but just 22.5% in the testing data, proving once again that the model may be overfitting.

To address overfitting, adjusting the hyperparameters like reducing the "max_depth", increasing the "min_child_weight", or adjusting the training data may be appropriate methods.

The result of important features selected by XGBoost Regressor is similar to Random Forest. As can be seen from Table 5, "Area" and "device_type" are the two most important features for the prediction of the model and "country" is still the least important one.

5 Conclusion

This study analyzed the impact of various factors on e-commerce delivery times using data from a ten-day period in 2021 in Hong Kong, employing both regression models and machine learning techniques such as Random Forest and XGBoost. Key factors such as geographical area and device type were identified as significant influencers of delivery speed, with additional insights regarding the effects of product subcategories, pricing strategies, and membership levels on delivery prioritization.

The findings can provide a few practical implications across the e-commerce spectrum. For e-commerce platforms, the data-driven insights enable targeted optimizations in delivery logistics, potentially leading to cost reductions and enhanced efficiency. This could involve refining delivery routes based on geographic analytics or tailoring promotional strategies to align with device usage patterns, thereby improving consumer engagement and loyalty. For suppliers, this research highlights the importance of strategic product prioritization and inventory management, which can help in planning production and aligning with platform algorithms to boost

product visibility and availability. Consumers stand to benefit from these improvements through more reliable and faster delivery services, alongside a more personalized and satisfying online shopping experience, ultimately leading to higher customer retention rates.

However, the study's scope was limited to just one platform in Hong Kong, which might not fully represent global e-commerce dynamics. Future research could broaden the geographic scope and integrate more detailed consumer data to enhance the universality and accuracy of the findings, further refining the strategies for global e-commerce operations.

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